

Hardware Abstraction Layer (HAL)

Applications need to talk to the system's hardware all the time. Application programmers should not be bothered with having to write new code for new hardware, so operating systems offer them the Hardware Abstraction Layer, or HAL. The HAL is a collection of *Device Drivers* which talk to the different kinds of hardware. They give applications a set of instructions which can be used to access this hardware. For example, there are different drivers for different sound-cards, but all of them give applications the same set of instructions to use. Applications use these instructions to create sound, and it is up to the driver to make the sound come out of the speakers. This way, all the application has to know is what functions it can use.

for other applications are then distributed among the remaining processors. In the second approach, *Symmetric Multi-processing*, all tasks, including those of the OS, are distributed evenly among all the processors.

1.2 The Heart, The Soul, The Kernel

Between applications and the hardware stands the *kernel*. It is the part of the operating system that performs the most basic and most critical functions that an OS is supposed to support. Its functions are:

- To make sure that all running applications (and the OS itself) get adequate CPU time to perform their operations.
- To manage the system memory and ensure that applications are getting what they need.
- To provide a *Hardware Abstraction Layer* (HAL), which hides the details of the hardware from applications, providing them instead with a set of instructions they can use to access the devices. This way, the application does not depend on which manufacturer the devices come from.
- To help applications talk to each other—this is called *Inter-Process Communication* (IPC).
- To manage files stored on the system's hard disk.